W6 PRACTICE

JS BASICS

At the end of this practice, you can

- ✓ Run JS code
- ✓ Create **variables** and **constants**
- ✓ Call and define **functions**
- ✓ Use JS **loops** and **conditions**
- ✓ Manipulate **arrays**, **objects**, **strings**, **Boolean** and **numbers**

Get ready before this practice!

- ✓ **Read** the following documents to understand JS syntax:

  https://cstart.mines.edu/web/Day2/2-JavaScriptBasicSyntax.pdf
  https://www.integral-domain.org/lwilliams/mis462/JavaScript.pdf

  You can also go further with the following books:
  https://www.gurukultti.org/admin/notice/javascript.pdf
  https://www.w3schools.com/js/default.asp

- ✓ **Complete the quiz** (*you can re-do it until you have 100% score*)

How to submit this practice?

- ✓ **Complete** this document
- ✓ Once finished, join this document to the MS Team assignment and **turn it in**



*3 WAYS TO **RUN JS CODE***

For beginners

To start with, you can just connect to an **online JavaScript editor**, such as this one:
https://playcode.io/javascript

For front-end ninjas

Chrome or any other **Web Browser** can execute JavaScript code while loading HTML

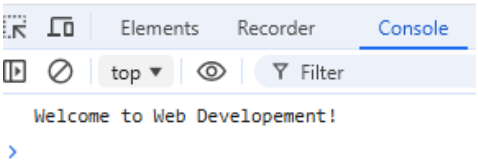
Just create a simple `index.html` file, that links to a `index.js` file:

```
<!DOCTYPE html>
<html>
<head>
    <title>Let s run JS on a Browser</title>
    <script src='index.js'></script>
</head>
<body>
</body>
</html>
```

Then just write some JS code, as example here, we print a message on the Browser console

```
// Example of JS code, printing on console
const courseName = "Web Developement";
console.log("Welcome to " + courseName + "!");
```

Finally open your index.html on a browser and check the console view



For back-end gurus

Node.js is also able to **execute JavaScript** code **outside a web browser.**
You will need first to install Node JS on your computer.

You can then just open a terminal on the folder containing your index.js file and run

```
node ./index.js
```

PART 1 - UNDERSTAND **JS SYNTAX**

Note: you can use the [C++ to JS converter] to compare C++ and JS syntax.



EXERCISE 1- TYPES, OUTPUTS

Analyze the differences between the provided C++ and JavaScript code.

C++	JS

`#include <iostream>`
`using namespace std;`
 	
`int main() {`
`  const int num = 5;`
`  for (int i = 0; i < num; i++) {`
`  cout << i << " ";`
`  }`
`  return 0;`
`}`
	`const num = 5;`
	`for (let i = 0; i < num; i++) {`
	`  console.log(i);`
	`}`

Q1 - What does the **const** key word mean in JS code?

const keyword mean that variable can't change the value.

Q2 - Why is it necessary **to specify the type** of variables in C++ but not in JavaScript?

Because C++ is static typing, it need to which compiler need to know data type first and how it manage the memory like int 4 bytes and char 1 byte. But in javascript they use dynamic typing which is checking in runtime, so it can hold different value at different time.

Q3- How to **print in the console** in JS?

console.log("message");

Q4- Is there any difference in the **loop syntax** between C++ and JS?

The only different in loop syntax is C++ need specific data type when initialize value i, but javascript we don't need to specify it, we just use "let" to initial i.

EXERCISE 2 - LOOPS, FUNCTIONS

C++

```cpp
#include <iostream>
using namespace std;

int calculateSum(int array[], int size) {
    int sum = 0;

    for (int i = 0; i < size; i++) {
      // Add here the calculation logic
    }
    return sum;
}

int main() {
    int arr[] = {1, 2, 3, 4, 5};
    cout << calculateSum(arr, 5);
    return 0;
}
```

JS

```js
function calculateSum(array) {
    let sum = 0;

    for (let i = 0; i < array.length; i++) {
       // Add here the calculation logic
    }
    return sum;
}

let arr = [1, 2, 3, 4, 5];
console.log(calculateSum(arr));
```

Q1 - Complete the given codes (`see comments`) to compute the sum of all elements in an array

Q2 – Why the function calculateSum in JS code **does not have the size** parameter?

Because JS have length attribute with array that count the size of array. So we don't have to use size parameter in that function.

EXERCISE 3 - CONDITIONS, EQUALITY

JS

```
function myFunction(min, max) {
  var result = "";
  for (let number = min; number <= max; number++) {
    if (number % 2 === 0) {
      result += number + " - ";
    }
  }
  return result;
} -
```

Q1 – Look at the above code

- Highlight all **variables in blue**
- Underline all **loops in red**
- Highlight all **conditions in green**

Q2 – What is the significance of the modulo operator % in these programs?

Modulo operator % in these programs when take number divide with two and return the remainder of that division.

Q3 – What is the difference between === and == in JS? *Highlight the right answer*

4 == 9	TRUE / FALSE
4 == 4	TRUE / FALSE
4 == ''4"	TRUE / FALSE
4 === ''4"	TRUE / FALSE

Q4 – What will this code will print on console?

```
console.log(myFunction(9, 14))
```

10 – 12 – 14 -

Q5 – What will this code will print on console?

```
console.log(myFunction(7, 3))
```

""

EXERCISE 4 – MEMORY ALLOCATION

Both codes are performing the same job:

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int size = 5;
    int* arr = new int[size];
    for (int i = 0; i < size; i++) {
        arr[i] = i * 2;
    }

    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    delete[] arr;
    return 0;
}
```

JS

```js
let size = 5;
let arr = [];
for (let i = 0; i < size; i++) {
    arr[i] = i * 2;
}

for (let i = 0; i < size; i++) {
    console.log(arr[i]);
}
```

Q1 – In both codes, are we using a **static** or a **dynamic** array? Explain why…

No, because in array in C++ don't have dynamic array. We have to copy the old array to the new array first then delete the old one if we want to change the size. But in JS we don't have to do like C++ because it use dynamic array.

Q2 – Explain why JavaScript **does not** need explicit **memory allocation** or **deallocation**, as C++ need it

Because C++ give the programmer direct control over system memory. But JS already design the for us. It will allocated behind the scenes when we declare the variable automatically.

*PART 2 - CODE **JS CHALLENGES***



Good job!

Now you should know the basic syntax of JavaScript!
Let's solve some problem now.

Each challenge is structured the same way:

- **Goal** What the function shall do
- **Inputs**: the function parameters
- **Output** the function return

As example, for the challenge 1, you will provide the following function:

```
function challenge1(width, height) {
    let rectangleString = '';
    // Your code
    return rectangleString;
}
```

CHALLENGE 1	EASY
Draw a rectangle in the console using stars	

INPUT	OUTPUT
width 3 height 4	*** *** *** ***
width 5 height 2	***** *****
width 5 height -2	

CHALLENGE 2	MEDIUM
Reverse an array	

INPUT	OUTPUT
array [14,15,16,20]	[20,16,15,14]
array [5,4,3,2,1]	[1,2,3,4,5]
array []	[]

Any help on arrays with JavaScript? Check here.

CHALLENGE 3	MEDIUM
Calculate the average grade of a list of students.	

INPUT	OUTPUT
array [85, 90, 78, 92]	86.25
array [10,20,30]	20
array []	0

CHALLENGE 4	MEDIUM

Write a function to count how many times a character appears in a string.

INPUT	OUTPUT
text "hello world" char = 'o'	2
text "aaa bbb a" char = 'a'	4
text "abc" char = 'd'	0

CHALLENGE 5	HARD

Count the number of words in a sentence

INPUT	OUTPUT
text "hello world"	2
text "this is the best day"	5
text "a bb ccc ddddddd e"	5

CHALLENGE 6	HARD

Simulate a voting system for three candidates (A / B/ C).
Count votes and declare a winner

INPUT	OUTPUT
votes ['A', 'B', 'A', 'C', 'A']	A is the winner
votes ['A', 'B', 'B', 'C', 'A']	A and B are both winners
votes []	There is not vote yet

Solution

Repository:

Challenge1

```
width 3 and height 4
***
***
***
***

width 5 and height 2
*****
*****

width 5 and height -2
```

Challenge2

```
before reverse  ▸ (4) [14, 15, 16, 20]          script.js:85
after reverse   ▸ (4) [20, 16, 15, 14]          script.js:87
before reverse  ▸ (5) [5, 4, 3, 2, 1]           script.js:88
after reverse   ▸ (5) [1, 2, 3, 4, 5]           script.js:90
before reverse  ▸ []                            script.js:91
after reverse   ▸ []                            script.js:93
› ctrl i to turn on code suggestions Don't show again
```

Challenge3

```
[85, 90, 78, 92]                                script.js:86
86.25
[10,20,30]                                      script.js:87
20
[]                                              script.js:88
0
```

Challenge4

```
hello world
find o
2
aaa bbb a
find a
4
abc
find d
0
```

Challenge5

```
hello world                                    script.js:86
2
this is the best day                           script.js:87
5
a bb ccc ddddddd e                             script.js:88
5
```

Challenge6

```
['A', 'B', 'A', 'C', 'A']                      script.js:87
A is the winner.                               script.js:73
['A', 'B', 'B', 'C', 'A']                      script.js:89
A and B are both winners.                      script.js:76
[]                                             script.js:91
There is no vote yet.                          script.js:58
```